Course Project Instruction

DSC 291: Trustworthy Machine Learning, SP'26
Instructor: Dr. Lily Weng

Instructions

- This is a group project with **4-5** students per group. You are free to make your own groups. A Piazza post is also launched to help you find teammates.
  - We will make arrangements to combine groups if necessary.
- The course project counts towards **40%** of your final grade.
- This project can be of two formats:
  - **Format 1. Tutorial on Research Reproduction** - Reproduce the results of relatively new research related to the course content
    - You will publish a tutorial on Medium on the research paper you reproduced, the topic should not already have many existing blogs on it.
    - This option has flexibility to conduct research e.g. design new experiments/methods and explore new research ideas.
  - **Format 2. Wikipedia Articles** - Contribute to Wikipedia articles that are closely related to the course materials (Robust ML and Interpretable ML)
    - You will create new article pages or new sections/subsections under existing pages and update outdated articles based on the topic/papers introduced in this course.
    - This option is similar to conducting a literature review.
- Students will submit a **project proposal** (approx 1-2 page) choosing their preferred format, its scope and expected implementation details.
- **Please read the rest of this document carefully** for project expectation/rubric, assignment deadlines, and grading format.

Timeline for Course Project

April 16, 2026 Thursday	Project Instruction released
May 7, 2026 Thursday, **12 PM**	Project Proposal Deadline
May 8, 2026 Friday	Proposal Revision Feedback released
May 14, 2026 Thursday, **5 PM**	Proposal Revision Deadline
June 9, 2026 Tuesday	Project presentations
June 9, 2026 Tuesday, **11:59 PM**	Slides, Report, Code, Rehearsal Feedback Form (Late policy: no late submission is allowed; late submission will be counted as zero grade) Do not email us for extensions!

[Project Overview:](#)

Project Proposal

- The proposals will be reviewed by the instructor and TA after submission
- If the proposals are in scope and only require minor revisions, **you do not need to resubmit the proposals if explicitly mentioned** by instructor or TA.
- Else, if proposals are deemed out of scope for the course or not feasible to complete within the course's timelines, revisions will need to be made after discussion with the instructor or TA and re-submitted as per the timeline.
- Project Proposal Submission details:
    - Submit your proposal pdf at gradescope: https://www.gradescope.com/courses/1287738/assignments/7990451
    - Please use the overleaf template provided below:
        - Research Reproduction: https://www.overleaf.com/read/dqppjwvrgscr#855353
        - Wikipedia Article: https://www.overleaf.com/read/ssmdksscpbsr#7f1eb0
    - If we do not receive your proposals by the deadline, you will be assigned paper(s) from the list at the end of the document or wikipedia article sections (with random teammates), and the 20% late penalty per day will still be applied.

For the **"Format 1: Tutorial on Research Reproduction"**

- Goal:
    - Reproduce research in an area relevant to the course material with potential opportunities in research exploration.
- Expectation:
    - Publish a blog/tutorial on Medium that contains steps on how to reproduce the results of a research paper, your analyses and inconsistencies/difficulties in reproduction
        - The work being reproduced must be from the list of papers at the Reference paper section at the end of this document.
        - You can also propose a research paper that is related to this course content and have been published at a top ML conference (e.g. NeurIPS, ICML, ICLR) or at OpenAI/Anthropic blogs between 2023-2026, however, you will need to justify in the proposal
            - Approval is not guaranteed and will depend on the justification
        - The tutorial should be accompanied with code and steps to reproduce the results. (Interactive code such as IPython Notebooks are preferred)
    - Class presentation (June 9)
        - This presentation should be rehearsed with another group to address initial concerns/doubts and will finally be presented in class.
- Grading:
    - **Proposal (15%)**
    - **Course Paper Presentation (55%):**
        - Presentation (30%)
        - Rehearsal with other group (15%)
        - Discussion (10%)

- **Report - Published Blog (20%)**
- **Code (10%)**
- **Bonus: Overall top 4 from both formats picked by class vote (10%)**

For the **“Format 2: Wikipedia Articles”**

- Goal:
  - Update outdated and create new Wikipedia material on research and developments related to the course content.
- Expectation:
  - Edit one or several of the below Wikipedia articles:
    - Explainable AI - Link
    - AI Safety - Link
    - Interpretable Machine Learning - Link
    - Adversarial Robustness - Link
    - Mechanistic Interpretability - Link
    - Jailbreak LLMs - New page or section needed
    - Concept Bottleneck Models - New page or section needed
    - Automated Interpretability - New page or section needed
  - If you find a different article not listed above that is relevant to course content and is missing information, you can justify it in the proposal and proceed after approval
    - Approval is not guaranteed and will depend on the justification
  - The article content should cover at least 10 research papers/blogs/articles and be approximately 2500 words long across 5 sections/subsections.
  - Class presentation (June 9)
    - A presentation of the article content should be prepared, but no rehearsal or discussion session is needed.
- Grading:
  - **Proposal (15%)**
  - **Course Paper Presentation (25%):**
    - Presentation (25%)
    - No rehearsal or discussion
  - **Report - Wikipedia Article (60%)**
  - **Bonus: Overall top 4 from both formats picked by class vote (10%)**

Proposal Rubric:

- **The proposal** is expected to be around **1-2 pages**
  - Group Members (4-5)
  - Format your group is planning to work on - Tutorial or Wikipedia
  - In case of Format 1: Tutorial
    - Research paper you are reproducing: please select 1 paper from the provided list at the Reference paper section (end of this doc)
    - A brief description of the work and topic
    - Results, tables or figures you are planning to reproduce

        - Additional experiments you plan to do (optional)
        - New ideas you plan to try (optional)
    - In case of Format 2: Wikipedia
        - Article you plan to work on
        - Sections you plan to add - you have to add 5 sections
        - Papers you plan to cover - you have to cover total 10 papers
        - New illustrations/figures/diagrams you plan to add (optional)

Project Rubric:

For the **"Format 1: Tutorial on Research Reproduction"**:

- **The presentation** is expected to be around **15** minutes (there might be some minor adjustment depending on the final number of teams).
    - Introduction to the topic and its importance/background (~1.5 mins)
        - What are the challenges?
        - What is the paper's goal and contributions?
    - Existing approaches to solve the problem and related works (~1.5 mins)
        - Introduce how prior work solve the problem and how are they different from this paper
        - Introduce the background required to understand this paper (particularly if the content is not covered in the class)
        - What is the novelty of the paper?
    - Key ideas and the main methods in the papers and comparison (~4 mins).
        - Start with the overall ideas / outlines / flow, and then get into the details of the ideas / methods / algorithms
        - Explain the key equations, and what's the intuition
    - Experiment results: setup, benchmark, baselines, evaluation metric and key results (~2 min)
        - Include key figures and key tables, make sure they are clear and easy to read
        - Explain the metrics, how to read the figures / tables, what are the results
    - Your reproduced results (~3 min)
        - Were you able to get the same reported results from the released code?
        - What's your reproduced result? Compare the numbers/plots you got v.s. the ones reported on the paper
    - Discussion & answer questions from the class (~3 min)
        - What do you like about the paper?
        - What do you not like about the paper?
        - What is the strength of the paper?
        - What is the weakness of the paper?
        - How does this paper compare with other work? Make a comparison table
        - What are the limitations of the paper? Has someone addressed them in the literature already?
    - Please use the google slides template that we provide, do not use other format (ppt, keynote, pdf, etc)

  Points to take care of:
    - Well designed slides
    - Properly timed presentation

    - Delivery of content
    - Covering the main sections above
- **Rehearsal:** Groups will be assigned in pairs and practice presentations (i.e. rehearsal) before the formal presentation in class. The purpose is to help you prepare the presentations better with the first audience's feedback.
    - Please refer for instructions: SP26 DSC 291 Rehearsal document
    - You are also welcome to present a draft of your presentation to TA and Prof. Weng during their office hours.
- **Report (Blogs)**: A web print of the blog in PDF format is to be submitted.
    - The blog content should approximately be a **10 min read** around **2500 words** (no more than 15 minutes)
    - The following sections are expected to be included in the blog (but flexible):
        - Introduction
        - Background and Related Work
        - Approach
        - Experiments
        - Setup, benchmarks, baselines, and evaluation metrics
        - Key results (tables/figures) and their analysis
        - Github Links and code snippets where applicable
        - New experiments or findings

For the **"Format 2: Wikipedia Articles"**:

- **The presentation** is expected to be around **5** minutes.
    - A brief summary of the topic that you are writing the article on (~1.5 mins)
    - Key ideas and the main contents that you are adding to the article (~3.5 mins).
    - Please use the google slides template that we provide, do not use other format (ppt, keynote, pdf, etc)

  Points to take care of:
    - Well designed slides
    - Properly timed presentation
    - Delivery of content
- **Report (Article)**: A web print of the article in PDF format is to be submitted.
    - The article content should cover content from at least 10 papers
    - Contribute to **5 sections/subsections** each with approximately 400-500 words (**Max 2500 words** cumulative)
    - We encourage you to submit early to Wikipedia after getting a draft approved by TA and Instructor.

Submission Format

1. Proposal

- Template
    - Research Reproduction: https://www.overleaf.com/read/dqppjwvrgscr#855353
    - Wikipedia Article: https://www.overleaf.com/read/ssmdksscpbsr#7f1eb0
- Note: Proposal has to be submitted as pdf on Gradescope

2. Presentation slides:

- Research reproduction template: SP 26 DSC 291 Project Template - Research Reproduction
    - Sample slides #1: SP 26 DSC 291 Example Certified Robustness
    - Sample slides #2: SP26 DSC 291 Example Interpretability
- Wikipedia article template: SP26 DSC 291 Project Template - Wikipedia Article
- Note: Presentation slides have to be submitted as pdf with google slides links on Gradescope

3. Rehearsal feedback form:
- Template: SP26 DSC 291 Rehearsal document
- Note: If you are doing Research reproduction format, please submit the rehearsal feedback forms as pdf on Gradescope.

4. Report & Code
- Research reproduction:
    - Please export your medium blog as pdf and submit the pdf and medium link on Gradescope
    - Please submit your code repo link on Gradescope
- Wikipedia article
    - Please export your Wikipedia article as pdf and submit the pdf and wikipedia link on Gradescope

Reference Papers

R1. Certified Robustness & Defenses & General Robustness

1. Smooth-Adv: Certified Adversarial Robustness via Randomized Smoothing (ICML19)
    a. Code: https://github.com/locuslab/smoothing
2. Boosting the Certified Robustness of L-infinity Distance Nets (ICLR22)
    a. Code: https://github.com/zbh2047/L_inf-dist-net-v2
3. (Certified!!) Adversarial Robustness for Free! (ICLR23)
    a. Code: https://github.com/ethz-spylab/diffusion_denoised_smoothing
4. Provably Robust Conformal Prediction with Improved Efficiency (ICLR24)
    a. Code: https://github.com/Trustworthy-ML-Lab/Provably-Robust-Conformal-Prediction
5. Breaking the Barrier: Enhanced Utility and Robustness in Smoothed DRL Agents (ICML24)
    a. Code: https://github.com/Trustworthy-ML-Lab/Robust_HighUtil_Smoothed_DRL
6. Corrupting Neuron Explanations of Deep Visual Features (ICCV23)
    a. Code: https://github.com/Trustworthy-ML-Lab/corrupting_neuron_explanations
7. Interpretability-Guided Test-Time Adversarial Defense (ECCV24)
    a. Code: https://github.com/Trustworthy-ML-Lab/Interpretability-Guided-Defense
8. Certified Robustness Under Bounded Levenshtein Distance (ICLR25)
    a. Code: https://github.com/LIONS-EPFL/LipsLev
9. GREAT Score: Global Robustness Evaluation of Adversarial Perturbation using Generative Models (NeurIPS24)

    a. Code: https://github.com/IBM/GREAT-Score
10. Your Diffusion Model is Secretly a Certifiably Robust Classifier (NeurIPS24)
    a. Code: https://github.com/huanranchen/NoisedDiffusionClassifiers

R2. Robustness in NLP
(a) Attacks

1. Jailbreaking Black Box Large Language Models in Twenty Queries (arXiv 2023)
    a. Code: https://github.com/patrickrchao/JailbreakingLLMs
2. Universal and Transferable Adversarial Attacks on Aligned Language Models (arXiv 2023)
    a. Code : https://github.com/llm-attacks/llm-attacks
3. Tree of Attacks: Jailbreaking Black-Box LLMs Automatically (arXiv 2023)
    a. Code: https://github.com/RICommunity/TAP?tab=readme-ov-file
4. Fast Adversarial Attacks on Language Models In One GPU Minute (ICML24)
    a. Code: https://github.com/vinusankars/BEAST
5. Iterative Self-Tuning LLMs for Enhanced Jailbreaking Capabilities (NAACL25)
    a. Code: https://github.com/SunChungEn/ADV-LLM
6. AdvPrompter: Fast Adaptive Adversarial Prompting for LLMs (ICML25)
    a. Code: https://github.com/facebookresearch/advprompter
7. SECA: Semantically Equivalent and Coherent Attacks for Eliciting LLM Hallucinations (NeurIPS'25)
    a. Code: https://github.com/Buyun-Liang/SECA

(b) Defense

8. Baseline defenses for adversarial attacks against aligned language models (arXiv 2023)
    a. Code: https://github.com/neelsjain/baseline-defenses
9. SmoothLLM: Defending Large Language Models Against Jailbreaking Attacks (arXiv 2023)
    a. Code: https://github.com/arobey1/smooth-llm
10. Improving Alignment and Robustness with Circuit Breakers (NeurIPS24)
    a. Code: github.com/GraySwanAI/circuit-breakers
11. Deliberative Alignment: Reasoning Enables Safer Language Models (arxiv 2024)

(c) Benchmarks

12. PromptBench: Towards Evaluating the Robustness of Large Language Models on Adversarial Prompts (arXiv 2023)
    a. Code: https://github.com/microsoft/promptbench
13. HarmBench: A Standardized Evaluation Framework for Automated Red Teaming and Robust Refusal (arXiv 2024)
    a. Code: https://github.com/centerforaisafety/HarmBench
14. JailbreakBench: An Open Robustness Benchmark for Jailbreaking Large Language Models (NeurIPS 2025)
    a. Code: https://github.com/JailbreakBench/jailbreakbench
    b. Dataset: https://huggingface.co/JailbreakBench

R3. Robustness in Agentic AI

1. InjecAgent: Benchmarking Indirect Prompt Injections in Tool-Integrated Large Language Model Agents (ACL 2024)
    a. Code :https://github.com/uiuc-kang-lab/InjecAgent
2. AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents (NeurIPS 2024)
    a. Website: https://agentdojo.spylab.ai/
3. AgentPoison: Red-teaming LLM Agents via Poisoning Memory or Knowledge Bases (NeurIPS 2024)
    a. Code: https://github.com/AI-secure/AgentPoison
4. AgentHarm: A Benchmark for Measuring Harmfulness of LLM Agents (ICLR 2025)
    a. Code: https://github.com/UKGovernmentBEIS/inspect_evals/tree/main/src/inspect_evals/agentharm
5. UDora: A Unified Red Teaming Framework against LLM Agents by Dynamically Hijacking Their Own Reasoning (ICML 2025)
    a. Code : https://github.com/AI-secure/UDora
6. Fully Autonomous AI Agents Should Not be Developed (arXiv 2025)
7. LLM-to-LLM Prompt Injection within Multi-Agent Systems (arXiv 2024)

I1. Neuron-level Interpretability techniques

1. Network Dissection: Quantifying Interpretability of Deep Visual Representations (CVPR17)
    - Code: http://netdissect.csail.mit.edu/
2. Natural Language Descriptions of Deep Visual Features (ICLR22)
    - Code: https://github.com/evandez/neuron-descriptions
3. Knowledge Neurons in Pretrained Transformers (ACL22)
    - Code: https://github.com/hunter-ddm/knowledge-neurons
4. CLIP-Dissect: Automatic Description of Neuron Representations in Deep Vision Networks (ICLR23)
    - Code: https://github.com/Trustworthy-ML-Lab/CLIP-dissect
5. Language models can explain neurons in language models (blog-paper)
    - Code: https://github.com/openai/automated-interpretability
6. The Importance of Prompt Tuning for Automated Neuron Explanations (NeurIPS’23W)
    - Code: https://github.com/Trustworthy-ML-Lab/Efficient-LLM-automated-interpretability
7. Linear Explanations for Individual Neurons (ICML24)
    - Code: https://github.com/Trustworthy-ML-Lab/Linear-Explanations
8. AND: Audio Network Dissection for Interpreting Deep Acoustic Models (ICML24)
    - Code: https://github.com/Trustworthy-ML-Lab/Audio_Network_Dissection
9. A Multimodal Automated Interpretability Agent (ICML24)
    - Code: https://github.com/multimodal-interpretability/maia
10. Interpreting Neurons in Deep Vision Networks with Language Models (TMLR 25)
    - Code: https://github.com/Trustworthy-ML-Lab/Describe-and-Dissect

11. Evaluating Neuron Explanations: A Unified Framework with Sanity Checks (ICML25)
    - Code: https://github.com/Trustworthy-ML-Lab/Neuron_Eval

I2. Model Editing, Debugging, Intervention

1. Concept-level Debugging of Part-Prototype Networks (ICLR 23)
    a. Code: https://github.com/abonte/protopdebug
2. Effective Skill Unlearning through Intervention and Abstention (NAACL 25)
    a. Code: https://github.com/Trustworthy-ML-Lab/effective_skill_unlearning
3. ThinkEdit: Interpretable Weight Editing to Mitigate Overly Short Thinking in Reasoning Models (EMNLP 25)
    a. Code: https://github.com/Trustworthy-ML-Lab/ThinkEdit
4. ODESteer: A Unified ODE-Based Steering Framework for LLM Alignment (ICLR 26)
    a. Code: https://github.com/ZhaoHongjue/odesteer
5. PaCE: Parsimonious Concept Engineering for Large Language Models (NeurIPS 24)
    a. Code: https://github.com/peterljq/Parsimonious-Concept-Engineering
6. Concept Lancet: Image Editing with Compositional Representation Transplant (CVPR'25)
    a. Code: https://github.com/peterljq/Concept-Lancet

I3. Interpretable Models by Design

1. Attention-based Interpretability with Concept Transformers (ICLR22)
    a. Code: https://github.com/ibm/concept_transformer
    b. efine
2. Interpretable by Design: Learning Predictors by Composing Interpretable Queries (IEEE TPAMI)
3. Post-hoc Concept Bottleneck Models (ICLR23)
    a. Code: https://github.com/mertyg/post-hoc-cbm
4. Label-free Concept Bottleneck Models (ICLR23)
    a. Code: https://github.com/Trustworthy-ML-Lab/Label-free-CBM
5. Variational Information Pursuit for Interpretable Predictions (ICLR 2023)
    a. Code:https://github.com/ryanchankh/VariationalInformationPursuit
6. Information Maximization Perspective of Orthogonal Matching Pursuit with Applications to Explainable AI (Neurips 2023)
    a. Code:https://github.com/r-zip/ip-omp
7. Bootstrapping Variational Information Pursuit with Large Language and Vision Models for Interpretable Image Classification (ICLR 2024)
    a. Code : https://openreview.net/pdf?id=9bmTbVaA2A
8. Contextual Knowledge Pursuit for Faithful Visual Synthesis (ECCV'24)
    a. Code: N/A
9. VLG-CBM: Training Concept Bottleneck Models with Vision-Language Guidance (NeurIPS 24)
10. Concept Bottleneck Language Models for Protein Design (arxiv 2024)
    a. Code: https://github.com/prescient-design/lobster

11. Concept Bottleneck Large Language Models (ICLR25)
    a. Code: https://github.com/Trustworthy-ML-Lab/CB-LLMs
12. Shedding Light on Time Series Classification using Interpretability Gated Networks (ICLR 25)
    a. Code: https://github.com/YunshiWen/InterpretGatedNetwork
13. InCoDe: Interpretable Compressed Descriptions For Image Generation (ICLR 2025)
    a. Code: https://github.com/ArmandCom/InCoDe
14. Interpretable Generative Models through Post-hoc Concept Bottlenecks (CVPR 25)
    a. Code: https://github.com/Trustworthy-ML-Lab/posthoc-generative-cbm
15. Show and Tell: Visually Explainable Deep Neural Nets via Spatially-Aware Concept Bottleneck Models (CVPR 25)
    a. Code: https://github.com/itaybenou/show-and-tell
16. ReFIne: A Framework for Trustworthy Large Reasoning Models with Reliability, Faithfulness, and Interpretability (arxiv 25)
    a. Code: https://github.com/Trustworthy-ML-Lab/Training_Trustworthy_LRM_with_R
17. Conformal Information Pursuit for Interactively Guiding Large Language Models (Neurips 2025)
    a. Code: https://github.com/ryanchankh/ConformalInformationPursuit
18. Learning Interpretable Queries for Explainable Image Classification with Information Pursuit (ICCV 2025)
19. Hierarchical Concept Embedding & Pursuit for Interpretable Image Classification (CVPR 2026)
    a. Code : https://github.com/nghiahhnguyen/hcep
20. Hierarchical Concept-based Interpretable Models (ICLR 2026)
    a. Code : https://github.com/OscarPi/cem-concept-discovery

I4. Interpretability in Agentic AI

1. AgentSHAP: Interpreting LLM Agent Tool Importance with Monte Carlo Shapley Value Estimation (arxiv 2025)
    a. Code:https://github.com/GenAISHAP/TokenSHAP
2. Interpreting Agentic Systems: Beyond Model Explanations to System-Level Accountability (arxiv 2026)
3. Transparency in Agentic AI: A Survey of Interpretability, Explainability, and Governance (arxiv 2026)
    a. Code:https://github.com/VectorInstitute/Agentic-Transparency